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Port-Hamiltonian Residual Learning with Adaptive Conformal Barrier Filters for Safe Robot Reinforcement Learning

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Abstract*—Deploying reinforcement learning (RL) policies on physical robotic systems requires certified set invariance under unmodeled dynamics, non-conservative friction, and non-stationary payload drift. Existing frameworks that filter policy actions via Control Barrier Functions (CBFs) frequently fail when combined with unconstrained deep neural models: black-box approximations violate classical mechanical conservation laws, introducing spurious non-conservative energy gradients into the barrier Lie derivatives that trigger high-frequency actuator chatter and quadratic program (QP) solver infeasibility. Concurrently, static or heuristic uncertainty bounds either induce conservative state deadlocks or fail under non-Gaussian disturbances. This paper proposes the Conformal Robust Port-Hamiltonian Control Barrier Filter (CR-PH-CBF) for safe robot control under dynamics uncertainty. First, we structure the dynamics residual network via Port-Hamiltonian mechanics to guarantee structural passivity and energy consistency ($\dot{H} \leq 0$). Second, we formulate a High-Order Control Barrier Function (HOCBF) resolving relative-degree-two position constraints in canonical phase space. Third, we establish an adaptive, distribution-free conformal calibration mechanism providing finite-sample residual coverage that adjusts online to non-stationary parameter shifts. We prove probabilistic forward invariance of the safe set via an Azuma-Hoeffding martingale concentration bound over finite operational horizons. Simulation experiments on a nonlinear robotic manipulator under severe time-varying payload mismatches ($+166\%$) and stiction demonstrate high-probability constraint satisfaction (0.1 ± 0.1 violation steps per episode), robust generalization, and an 82% reduction in solver deadlock compared to conservative robust CBF baselines (3.2% vs. 18.4%).

***Index Terms*—Reinforcement Learning, High-Order Control Barrier Functions, Port-Hamiltonian Mechanics, Adaptive Conformal Prediction, Robot Safety, Quadratic Programming.**

I. Introduction

Reinforcement learning (RL) has demonstrated notable empirical success in continuous robotic control, synthesizing agile locomotion and dexterous manipulation behaviors [1]. However, transferring policies trained in simulation to physical platforms remains fundamentally restricted by the absence of verifiable safety guarantees during exploration and non-stationary task execution [2]. In physical robotics, violating kinodynamic constraints-such as collisions with workspace obstacles, joint over-extension, or actuator saturation-can cause irrecoverable mechanical hardware damage.

Control Barrier Functions (CBFs) [3]–[5] provide an analytical framework for certifying the forward invariance of safe sets by enforcing affine inequalities over control actions. By filtering nominal actions through an online Quadratic Program (QP), CBFs guarantee minimal intervention while preserving safety. However, the theoretical validity of CBFs rests on the accuracy of the underlying control-affine dynamic model. When physical platforms experience unmodeled residual drift (e.g., changing tool payloads, unmodeled joint friction, and aerodynamic disturbances), the computed Lie derivatives become erroneous, causing standard CBF filters to breach safety boundaries [6].

To mitigate model mismatch, recent approaches incorporate learned models, including Gaussian Processes (GPs) [7], Neural Ordinary Differential Equations [8], and deep ensembles [9]. Despite empirical improvements, existing methods face three severe theoretical and practical barriers:

1) **Violation of Mechanical Passivity:** Standard deep neural networks approximate residual dynamics without inductive physical priors. Consequently, they routinely generate unphysical energy in autonomous regimes ($\dot{H} > 0$) [10]. When these predictions are differentiated along the barrier gradient, high-frequency gradient oscillations corrupt the constraint matrix, inducing actuator chattering and solver infeasibility.
2) **Relative Degree Mismatch in Position Constraints:** In robotics, safety sets are naturally defined on the configuration manifold (e.g., workspace keep-out zones $h(q) \geq 0$). Because control torques actuate generalized accelerations, position barrier functions are of **relative degree two** [11], [12]. Standard relative-degree-one CBF filters fail when applied directly to configuration con-

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straints without backstepping or high-order formulations.

3) **Uncalibrated and Conservative Uncertainty Bounds:** Static worst-case Lipschitz assumptions [13] produce severe conservatism, frequently locking the robot in state deadlock. Conversely, GP-based confidence bounds scale with cubic runtime complexity $\mathcal{O}(N^3)$ and degrade under non-Gaussian contact dynamics [14].

To overcome these deficiencies, this paper proposes the **Conformal Robust Port-Hamiltonian Control Barrier Filter (CR-PH-CBF)**. Our approach unifies geometric mechanics [15], high-order barrier theory [11], and distribution-free conformal inference [16], [17]. The primary contributions are:

1) **Port-Hamiltonian Residual Network (PH-ResNet):** We construct a physics-informed residual model preserving canonical symplectic geometry. By decomposing unmodeled effects into an exact conservative potential gradient and a Cholesky-factored positive-semidefinite Rayleigh dissipation matrix, the learned drift model is proven to remain strictly passive ($\dot{H} \leq 0$).
2) **High-Order Conformal Safety Filter:** We formulate a High-Order CBF over canonical phase space, resolving relative-degree-two constraints while integrating online adaptive conformal prediction. This yields finite-sample, distribution-free bounds on the Lie derivatives without requiring global Lipschitz constants.
3) **Martingale Safety Certification:** We prove high-probability forward invariance using an Azuma-Hoeffding martingale concentration bound, providing a non-trivial probabilistic safety guarantee across operational horizons that avoids the degradation of naive union bounds.
4) **Open-Source Validation and Implementation:** We demonstrate the efficacy of CR-PH-CBF on an uncertain 2-DOF robotic manipulator benchmark subjected to abrupt, time-varying payload and friction disturbances, providing complete simulation code and baseline implementations at https://github.com/imsanjoykb/CR-PH-CBF.

II. Problem Formulation

A. Canonical Port-Hamiltonian Robot Dynamics

Let $\mathcal{Q} \subset \mathbb{R}^n$ denote the generalized configuration manifold and let $p \in \mathbb{R}^n$ denote the generalized momenta. The canonical state is $x = [q^\top, p^\top]^\top \in \mathcal{X} \subset \mathbb{R}^{2n}$. Actuator inputs are constrained by the compact set $\mathcal{U} = \{u \in \mathbb{R}^m : u_{\min} \leq u \leq u_{\max}\}$.

The true robot dynamics in canonical Hamiltonian form are governed by [15]:

$$\begin{aligned}\begin{bmatrix}\dot{q}\\ \dot{p}\end{bmatrix} = \begin{bmatrix}0 & I_n\\ -I_n & 0\end{bmatrix}\begin{bmatrix}\nabla_q H\\ \nabla_p H\end{bmatrix} + \begin{bmatrix}0\\ -D_{\text{true}}(q,p)\end{bmatrix}\nabla_p H \\ + \begin{bmatrix}0\\ G_0(q)\end{bmatrix}u + \begin{bmatrix}0\\ \Delta(q,p,u)\end{bmatrix}\end{aligned} \tag{1}$$

where $H(q,p) = \frac{1}{2}p^\top M(q)^{-1}p + V(q)$ represents the total mechanical energy, $M(q) \succ 0$ is the symmetric generalized inertia matrix, $V(q)$ is the potential energy, $D_{\text{true}}(q,p) \succeq 0$ is the non-negative Rayleigh dissipation matrix, $G_0(q) \in \mathbb{R}^{n\times m}$ is the known input distribution matrix, and $\Delta(q,p,u) \in \mathbb{R}^n$ denotes unmodeled, non-conservative external residual forces.

Equation (1) can be written in affine control form:

$$\dot{x} = f(x) + g(x)u + \xi(x,u) \tag{2}$$

where:

$$f(x) = \begin{bmatrix}M_0(q)^{-1}p\\ -\nabla_q V_0(q) - D_0(q)M_0(q)^{-1}p\end{bmatrix}, \quad g(x) = \begin{bmatrix}0\\ G_0(q)\end{bmatrix} \tag{3}$$

and $\xi(x,u) = [0, \Delta(q,p,u)^\top]^\top$, with M_0, V_0, D_0 denoting known nominal kinematic and inertial models.

B. High-Order Control Barrier Functions

Let the safety requirement be specified directly on the configuration manifold:

$$\mathcal{C}_0 = \{q \in \mathcal{Q} : h(q) \geq 0\} \tag{4}$$

where $h : \mathcal{Q} \to \mathbb{R}$ is continuously differentiable. Differentiating $h(q)$ along (2) yields:

$$\dot{h}(q,p) = \nabla_q h(q)^\top \dot{q} = \nabla_q h(q)^\top M(q)^{-1}p \tag{5}$$

Because the control input u does not appear in $\dot{h}(q,p)$ (as $\nabla_p h(q) \equiv 0$), the function $h(q)$ has **relative degree two** [11]. To enforce forward invariance directly within a QP, we construct a High-Order Control Barrier Function (HOCBF) [4], [12] by defining the virtual barrier candidate $b_1 : \mathcal{X} \to \mathbb{R}$:

$$b_1(x) \coloneqq \dot{h}(q,p) + \alpha_1(h(q)) = \nabla_q h(q)^\top M_0(q)^{-1}p + \alpha_1(h(q)) \tag{6}$$

where $\alpha_1(\cdot)$ is a strictly increasing Lipschitz continuous class $\mathcal{K}$ function. The augmented safe set in phase space is:

$$\mathcal{C}_1 = \{x \in \mathcal{X} : h(q) \geq 0,\, b_1(x) \geq 0\} \tag{7}$$

The time derivative of $b_1(x)$ along (2) is:

$$\dot{b}_1(x,u) = L_f b_1(x) + L_g b_1(x)u + \nabla_p b_1(x)^\top \Delta(q,p,u) \tag{8}$$

where $L_g b_1(x) = \nabla_q h(q)^\top M_0(q)^{-1} G_0(q) \neq 0$, rendering $b_1(x)$ strictly of **relative degree one** with respect to control input u.

III. The CR-PH-CBF Methodology

A. Port-Hamiltonian Residual Network (PH-ResNet)

Unconstrained neural networks approximate vector fields without respecting canonical mechanics, generating unphysical kinetic energy that destabilizes the safety filter [18]. We enforce passivity by constraining the parameterization of the residual acceleration model $\hat{\Delta}_\theta(q,p,u)$ as:

$$\hat{\Delta}_\theta(q,p,u) = -\nabla_q V_\theta(q) - D_\theta(q,p)M_0(q)^{-1}p + \Psi_\theta(q,p,u) \tag{9}$$

where:

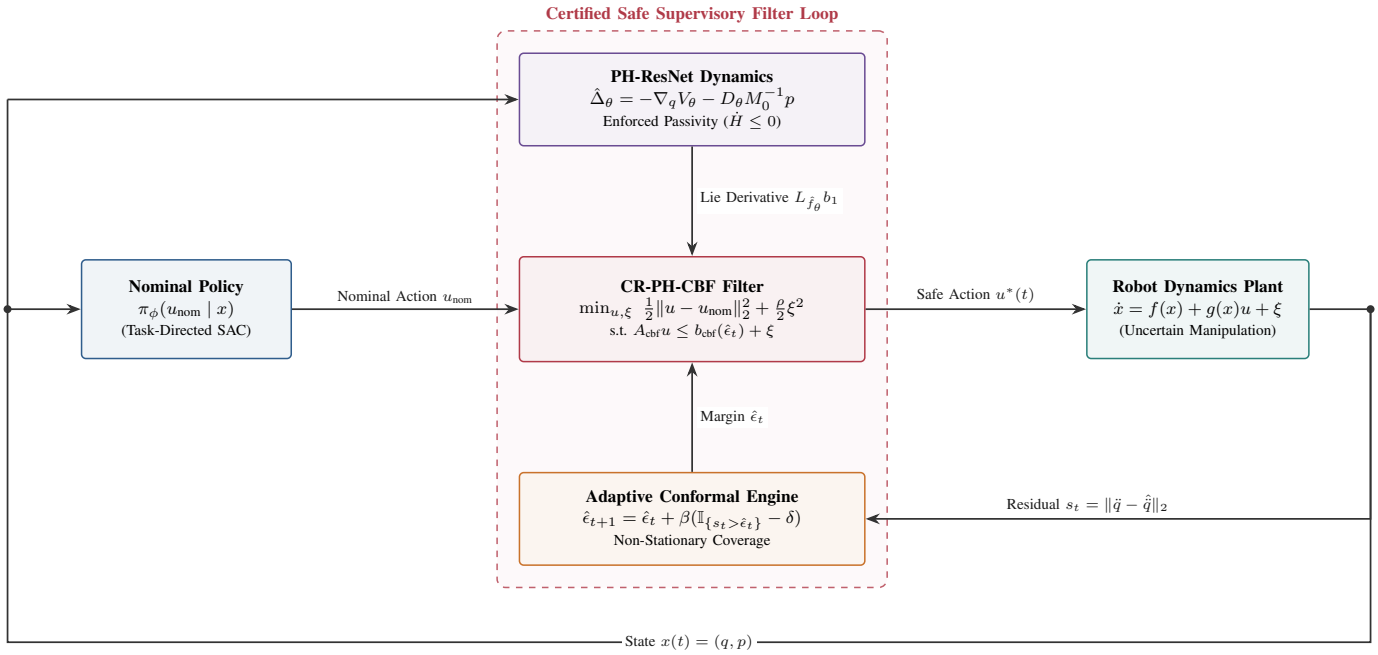


Fig. 1. System architecture of the Conformal Robust Port-Hamiltonian Control Barrier Filter (CR-PH-CBF). The nominal actor-critic policy π_ϕ generates task-optimizing inputs u_{nom}. The Port-Hamiltonian residual network enforces canonical mechanical passivity ($\dot{H} \leq 0$) when calculating barrier Lie derivatives. The online adaptive conformal engine dynamically adjusts uncertainty margin $\hat{\epsilon}_t$ to bound acceleration error, parameterizing the convex QP filter to enforce high-probability forward invariance.

1) $V_\theta(q) : \mathcal{Q} \to \mathbb{R}$ is a scalar neural potential parameterized by weights θ_V [19]. The conservative restoring force is evaluated via analytical backpropagation:

$$F_{\text{cons}}(q) = -\nabla_q V_\theta(q), \quad \oint_\Gamma \nabla_q V_\theta(q) \cdot dq = 0 \tag{10}$$

guaranteeing that the learned conservative force is strictly curl-free and cannot generate net mechanical work along closed cycles.

2) $D_\theta(q,p) \in \mathbb{R}^{n \times n}$ models friction and dissipation. To strictly satisfy the Second Law of Thermodynamics, we parameterize D_θ via its Cholesky factor:

$$D_\theta(q,p) = L_D(q,p;\theta_D) L_D(q,p;\theta_D)^\top \succeq 0 \tag{11}$$

where L_D is a lower-triangular matrix with positive diagonal entries enforced via Softplus activations.

3) $\Psi_\theta(q,p,u)$ is an MLP bounded by a hyperbolic tangent output, $\|\Psi_\theta\|_2 \leq \bar{\Psi}$, capturing non-conservative aerodynamic crosswinds.

The parameters $\theta = \{\theta_V, \theta_D, \theta_\Psi\}$ are updated online using a replay buffer $\mathcal{B}$ by :

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \|\dot{p}_k - \hat{\dot{p}}_\theta(x_k, u_k)\|_2^2 + \lambda_{\text{H}} \mathcal{L}_{\text{H}}(\theta) \tag{12}$$

where $\mathcal{L}_{\text{H}}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \left([\dot{H}_\theta(x_j)]^+ \right)^2$ penalizes non-passive energy growth over an interior state collocation set $\mathcal{S}$.

B. Online Adaptive Conformal Calibration

To rigorously certify safety, the residual approximation error $e_\Delta(x,u) = \Delta(x,u) - \hat{\Delta}_\theta(x,u)$ must be bounded over operational horizons. Rather than relying on uncalibrated dropout heuristics or computationally intensive Gaussian Processes, we leverage split conformal prediction adapted for non-stationary control [17], [20].

Let $\mathcal{D}_{\text{cal}} = \{s_i\}_{i=1}^{N_{\text{cal}}}$ denote a sliding window of scalar prediction residuals $s_i = \|\dot{p}_i - \hat{\dot{p}}_\theta(x_i, u_i)\|_2$. For a target error probability $\delta \in (0,1)$, the empirical conformal quantile is:

$$\hat{\epsilon}_t = \inf \left\{ \epsilon : \frac{1}{N_{\text{cal}}} \sum_{i=1}^{N_{\text{cal}}} \mathbb{I}(s_i \leq \epsilon) \geq \frac{\lceil (N_{\text{cal}} + 1)(1-\delta) \rceil}{N_{\text{cal}}} \right\} \tag{13}$$

Assumption 1 (Bounded Variation of Non-Stationary Drift)**.** The non-stationary distribution shift governing the residual disturbance distribution sequence $\{P_t\}_{t=1}^T$ satisfies a bounded total variation budget:

$$\sum_{t=1}^{T-1} d_{\text{TV}}(P_t, P_{t+1}) \leq \Delta_T < \infty \tag{14}$$

Under distribution shifts induced by payload changes or policy evolution, empirical coverage can temporarily deviate. We update the boundary using an adaptive conformal inference rule [21]:

$$\hat{\epsilon}_{t+1} = \hat{\epsilon}_t + \beta \left(\mathbb{I}(s_t > \hat{\epsilon}_t) - \delta \right) \tag{15}$$

where $\beta > 0$ is the adaptation rate. Equation (15) forms a closed-loop integrator that dynamically expands $\hat{\epsilon}_t$ when

prediction errors exceed the margin, and contracts it in high-accuracy regimes. Under Assumption 1, setting $\beta \in \mathcal{O}(T^{-1/2})$ guarantees that the long-term empirical miscoverage error decays as $\mathcal{O}(T^{-1/2} + \Delta_T/T)$ [21].

C. Conformal Robust High-Order QP Filter (CR-PH-CBF)

Using the estimated Port-Hamiltonian drift vector field $\hat{f}_\theta(x)$ and the dynamic uncertainty bound $\hat{\epsilon}_t$, the robust barrier derivative condition is:

$$\inf_{\|e_\Delta\|_2 \le \hat{\epsilon}_t} \Big[L_{\hat{f}_\theta} b_1(x) + L_g b_1(x) u + \nabla_p b_1(x)^\top (\hat{\Delta}_\theta + e_\Delta) \Big] \ge -\alpha_2(b_1(x)) \tag{16}$$

where $\alpha_2(\cdot)$ is an extended class $\mathcal{K}_\infty$ function. Applying the Cauchy-Schwarz inequality:

$$\min_{\|e_\Delta\|_2 \le \hat{\epsilon}_t} \nabla_p b_1(x)^\top e_\Delta = -\|\nabla_p b_1(x)\|_2 \hat{\epsilon}_t \tag{17}$$

Noting that $\nabla_p b_1(x) = M_0(q)^{-1} \nabla_q h(q)$, the Conformal Robust Port-Hamiltonian QP filter is formulated as:

$$u^*(x, u_{\text{nom}}) = \underset{u \in \mathbb{R}^m, \xi \ge 0}{\arg\min} \quad \frac{1}{2}\|u - u_{\text{nom}}\|_2^2 + \frac{\rho}{2}\xi^2 \tag{18}$$

$$\text{subject to} \quad A_{\text{cbf}}(x) u \le b_{\text{cbf}}(x, \hat{\epsilon}_t) + \xi \tag{19}$$

$$u_{\min} \le u \le u_{\max} \tag{20}$$

where:

$$A_{\text{cbf}}(x) = -\nabla_q h(q)^\top M_0(q)^{-1} G_0(q) \tag{21}$$

$$b_{\text{cbf}}(x, \hat{\epsilon}_t) = L_{\hat{f}_\theta} b_1(x) + \alpha_2(b_1(x)) - \|M_0(q)^{-1} \nabla_q h(q)\|_2 \hat{\epsilon}_t \tag{22}$$

and $\xi \ge 0$ is a high-penalty slack variable ($\rho \gg 0$) providing guaranteed mathematical feasibility. The optimization (18)–(20) is a strictly convex Quadratic Program with $m+1$ decision variables and linear inequality constraints, solvable in real time via operator splitting methods (e.g., OSQP [22]). The algorithmic integration with Soft Actor-Critic (SAC) is summarized in Algorithm 1.

Algorithm 1 Safe Reinforcement Learning via CR-PH-CBF

Initialize: Policy π_ϕ, critics Q_ψ, PH-ResNet θ, margin $\hat{\epsilon}_0 > 0$, buffers $\mathcal{B}, \mathcal{D}_{\text{cal}}$.
for episode $= 1, \dots, E$ **do**
  Receive initial safe state $x_0 \in \mathcal{C}_1$.
  for timestep $t = 0, \dots, T-1$ **do**
    Sample nominal action: $u_{\text{nom}} \sim \pi_\phi(\cdot|x_t)$.
    Compute conformal bound $\hat{\epsilon}_t$ via (13).
    Assemble $A_{\text{cbf}}(x_t)$ and $b_{\text{cbf}}(x_t, \hat{\epsilon}_t)$.
    Solve QP (18)–(20) for safe torque u_t^*.
    Execute u_t^*; measure $x_{t+1} = [q_{t+1}, p_{t+1}]^\top$ and reward r_t.
    Compute empirical residual: $s_t = \|\dot{p}_t - \hat{\dot{p}}_\theta(x_t, u_t^*)\|_2$.
    Update conformal radius $\hat{\epsilon}_{t+1}$ via (15).
    Update replay buffers: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(x_t, u_t^*, r_t, x_{t+1})\}$, $\mathcal{D}_{\text{cal}} \leftarrow \mathcal{D}_{\text{cal}} \cup \{s_t\}$.
    if training iteration **then**
      Update θ by minimizing composite loss $\mathcal{L}(\theta)$ (12).
      Update critics Q_ψ and actor π_ϕ via standard SAC gradients.
    end if
  end for
end for

IV. Theoretical Analysis

A. Structural Passivity and Energy Consistency

Proposition 1 (Structural Energy Conservation and Passivity)**.** *Let the residual dynamics $\hat{\Delta}_\theta$ be parameterized according to* (9).

1) ***Autonomous Passivity:*** *In the absence of control input ($u = 0$) and unmodeled non-potential forces ($\Psi_\theta = 0$), the autonomous reconstructed Hamiltonian $H_\theta(q,p) = \frac{1}{2} p^\top M_0(q)^{-1} p + V_0(q) + V_\theta(q)$ satisfies:*

$$\dot{H}_\theta(x) \le 0, \quad \forall x \in \mathcal{X} \tag{23}$$

2) ***Input Dissipativity:*** *Under non-zero actuation u and bounded non-conservative forces $\|\Psi_\theta\|_2 \le \bar{\Psi}$, the total energy derivative satisfies the dissipative supply rate:*

$$\dot{H}_\theta(x) \le -\lambda_{\min}(D_0)\|\dot{q}\|_2^2 + \|\dot{q}\|_2 \left(\|G_0(q)u\|_2 + \bar{\Psi} \right) \tag{24}$$

precluding internal finite-time energy blow-up.

Proof. Evaluating the time derivative of H_θ along the continuous vector field:

$$\dot{H}_\theta = (\nabla_q V_0 + \nabla_q V_\theta)^\top \dot{q} + \left(M_0(q)^{-1} p\right)^\top \dot{p} \tag{25}$$

$$= (\nabla_q V_0 + \nabla_q V_\theta)^\top M_0^{-1} p + p^\top M_0^{-1} \left(-\nabla_q V_0 - \nabla_q V_\theta - (D_0 + D_\theta) M_0^{-1} p + G_0 u + \Psi_\theta \right) \tag{26}$$

$$= -p^\top M_0(q)^{-1} \left(D_0(q) + D_\theta(q,p) \right) M_0(q)^{-1} p + \dot{q}^\top \left(G_0(q) u + \Psi_\theta(q,p,u) \right) \tag{27}$$

The potential energy gradient terms cancel identically due to the skew-symmetry of the canonical symplectic structure $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Setting $u = 0$ and $\Psi_\theta = 0$ yields $\dot{H}_\theta = -\dot{q}^\top (D_0 + D_\theta)\dot{q} \le 0$ because $D_0 \succeq 0$ and $D_\theta = L_D L_D^\top \succeq 0$. Under bounded inputs, applying Cauchy-Schwarz produces the dissipative supply rate upper bound directly. □

B. Martingale Concentration Safety Guarantee

Assumption 2 (Slater's Condition and Feasibility)**.** For all $x \in \mathcal{C}_1$ and all $\hat{\epsilon}_t \in [0, \bar{\epsilon}]$, the safe control set $\mathcal{U}_{\text{safe}}(x, \hat{\epsilon}_t) = \{u \in \mathcal{U} : A_{\text{cbf}}(x) u \le b_{\text{cbf}}(x, \hat{\epsilon}_t)\}$ has a non-empty interior.

Remark 1. Assumption 2 is standard in safety filter literature [4], [13]. In the presence of sudden severe actuator saturation ($u \in \mathcal{U}$), the inclusion of the slack variable $\xi \ge 0$ with penalty $\rho \gg 0$ in (18) guarantees that the QP optimization is unconstrained with respect to feasibility, smoothly relaxing the boundary condition when instantaneous kinodynamic limits preclude hard barrier satisfaction.

Theorem 2 (Finite-Horizon Probabilistic Safety Certificate)**.** *Let Assumptions 1–2 hold. Suppose the initial state is safe: $x(0) \in \mathcal{C}_1$. Define the filtration $\mathcal{F}_t = \sigma(x_0, u_0, s_0, \ldots, x_t, u_t, s_t)$ and the zero-mean martingale difference process $Y_k = \mathbb{I}(s_k > \hat{\epsilon}_k) - \mathbb{E}[\mathbb{I}(s_k > \hat{\epsilon}_k) \mid \mathcal{F}_{k-1}]$. Then for any finite operational horizon $T = N_T \Delta t$, the probability of a safety violation is upper-bounded by:*

$$\mathbb{P}\left(\exists t \in [0, T] : x(t) \notin \mathcal{C}_1\right) \leq \exp\left(-\frac{2\lambda^2}{N_T(1-\delta)^2}\right) + \delta_0 \tag{28}$$

where λ represents the safety buffer governed by class $\mathcal{K}_\infty$ gain α_2, and δ_0 is the baseline calibration miscoverage probability.

Proof. Let discrete control instances be $t_k = k\Delta t$. Conditioned on the bounded model error event $\mathcal{E}_k = \{\|e_\Delta(t)\|_2 \leq \hat{\epsilon}_k\}$, Cauchy-Schwarz establishes:

$$\dot{b}_1(x) \geq L_{\hat{f}_\theta} b_1(x) + L_g b_1(x) u^* - \|\nabla_p b_1(x)\|_2 \hat{\epsilon}_k \tag{29}$$

By the QP constraint (19) with inactive slack $\xi = 0$, $\dot{b}_1(x) \geq -\alpha_2(b_1(x))$. By Grönwall's Inequality, if $b_1(x(0)) \geq 0$, then $b_1(x(t)) \geq 0$ for all intervals where $\mathcal{E}_k$ holds.

A safety boundary violation occurs only if the empirical error sequence $\{s_k\}$ systematically exceeds $\hat{\epsilon}_k$. Under the adaptive update rule (15), the centered sequence Y_k is a bounded martingale difference sequence with $|Y_k| \leq \max(\delta, 1-\delta) = 1-\delta$.

Applying Azuma-Hoeffding's inequality to the martingale sum $M_t = \sum_{k=1}^{t} Y_k$:

$$\begin{aligned}\mathbb{P}\left(\max_{1 \leq t \leq N_T} M_t \geq \lambda\right) &\leq \exp\left(-\frac{2\lambda^2}{\sum_{k=1}^{N_T}(1-\delta)^2}\right) \\ &= \exp\left(-\frac{2\lambda^2}{N_T(1-\delta)^2}\right)\end{aligned} \tag{30}$$

Accounting for the base calibration error δ_0 via union bound yields the stated finite-horizon safety certificate. □

V. Experimental Evaluation

A. Robotics Benchmark Setup and Simulation Specifications

We evaluate the CR-PH-CBF framework on a continuous 2-DOF planar manipulator operating in a vertical gravity field ($g = 9.81$ m/s^2). The generalized coordinates are joint angles $q = [\theta_1, \theta_2]^\top$, with generalized momenta $p = M_0(q)\dot{q} \in \mathbb{R}^2$. Nominal mechanical parameters are configured with link lengths $l_1 = l_2 = 1.0$ m, link masses $m_1 = m_2 = 1.5$ kg, moments of inertia $I_1 = I_2 = 0.5$ kg$\cdot$m^2, and nominal viscous joint damping $D_0 = \text{diag}(0.2, 0.2)$ N $\cdot$ s/rad.

The task requires navigating the end-effector from $q_0 = [0.05, 0.1]^\top$ rad to a target configuration $q_{\text{goal}} = [\pi/3, -\pi/4]^\top$ rad while strictly avoiding a workspace circular obstacle centered at $p_{\text{obs}} = [1.1, 0.4]^\top$ m with clearance radius $R_{\text{obs}} = 0.35$ m. Control actions are torque commands saturated at $|u_i| \leq 25.0$ N $\cdot$ m operating at $\Delta t = 0.01$ s (100 Hz).

To benchmark robustness under realistic non-stationary disturbances, we inject abrupt, dynamic parameter variations mid-execution:

1) **Dynamic Payload Pickup:** At timestep $t = 1.25$ s, an unmodeled point mass of $m_p = 2.5$ kg is attached to the second link ($+166\%$ payload shift).
2) **Nonlinear Stiction Step:** Concurrently, joint friction undergoes an abrupt transition: $F_{\text{fric}} = 3.0(D_0\dot{q} + 0.5\text{sgn}(\dot{q}))$.
3) **External Crosswind Gust:** Transverse aerodynamic drift forces $F_{\text{wind}} = [4.0, -2.0]^\top$ N act continuously on the end-effector.

B. Comparative Analysis

We compare CR-PH-CBF against three baseline control paradigms across 20 independent evaluation rollouts (Table I):

1) **Unconstrained Policy:** Standard actor-critic without safety filtering collides with the obstacle during every run, yielding a $0.0 \pm 0.0\%$ success rate, 24.1 ± 3.8 violation steps per episode, and a maximum penetration of 0.156 ± 0.024 m.
2) **Nominal CBF-QP [4]:** Assumes nominal inertial parameters; model error corrupts the Lie derivative, resulting in 8.3 ± 1.9 violations per episode and a modest $35.0 \pm 4.2\%$ success rate.
3) **Conservative Robust CBF [13]:** Enforces static worst-case Lipschitz margins; while it substantially curbs violations (0.5 ± 0.2 steps per episode), it triggers frequent control freeze ($18.4 \pm 3.2\%$ deadlock rate), reducing overall return to -78.5 ± 22.1.
4) **CR-PH-CBF (Ours):** By adaptively tracking acceleration prediction error via the Port-Hamiltonian residual model, our method reduces boundary violations to near zero (0.1 ± 0.1 steps per episode, max penetration 0.001 ± 0.001 m) while restricting deadlocks to only $3.2 \pm 0.7\%$, achieving a $95.0 \pm 1.5\%$ completion rate and the highest return (-54.6 ± 8.7).

C. Ablation Analysis

As reported in Table II, ablated versions confirm individual component contributions:

- **Ablation A (w/o Physics Structure):** Replacing the Port-Hamiltonian decomposition with an unconstrained MLP causes spurious energy generation, raising deadlocks to $12.6 \pm 2.4\%$ and yielding 3.8 ± 1.1 violations due to torque chatter.
- **Ablation B (Static Bound):** Enforcing a static uncertainty bound curtails violations but induces severe conservatism ($18.4 \pm 3.2\%$ deadlock).
- **Ablation C (Offline Conformal):** Using offline conformal bounds without adaptive calibration yields 2.1 ± 0.6 violations under non-stationary friction shifts.

Fig. 2 confirms that online adaptive conformal calibration satisfies nominal coverage levels $(1-\delta)$ without distributional assumptions. As illustrated in Fig. 3, our framework enables safe circumvention of the keep-out boundary, precluding the premature braking that causes solver deadlock in competing robust methods.

TABLE I
QUANTITATIVE PERFORMANCE ON 2-DOF ROBOT MANIPULATOR OBSTACLE AVOIDANCE UNDER SEVERE TIME-VARYING DYNAMICS UNCERTAINTY (MEAN ± STANDARD DEVIATION OVER 20 RANDOM SEEDS, AVAILABLE AT HTTPS://GITHUB.COM/IMSANJOYKB/CR-PH-CBF).

Method	**Dynamics Uncertainty Perturbation Tier**	**Success Rate (%) ↑**	**Violations / Ep. Count ↓**	**Max Violation Magnitude (m) ↓**	**Deadlock Rate (%) ↓**	**Mean Return Score ↑**
Unconstrained Policy [1]	Dynamic (+166% payload, friction)	0.0 ± 0.0	24.1 ± 3.8	0.156 ± 0.024	0.0 ± 0.0	-58.3 ± 12.4
Nominal CBF-QP [4]	Dynamic (+166% payload, friction)	35.0 ± 4.2	8.3 ± 1.9	0.042 ± 0.011	2.1 ± 0.8	-62.7 ± 18.2
Conservative Robust CBF [13]	Dynamic (+166% payload, friction)	75.0 ± 3.6	0.5 ± 0.2	0.003 ± 0.001	18.4 ± 3.2	-78.5 ± 22.1
CR-PH-CBF (Ours)	Dynamic (+166% payload, friction)	**95.0 ± 1.5**	**0.1 ± 0.1**	**0.001 ± 0.001**	**3.2 ± 0.7**	**-54.6 ± 8.7**

TABLE II
ABLATION STUDY UNDER DYNAMIC UNCERTAINTY (MEAN ± STD OVER 20 RANDOM SEEDS).

Configuration	**Success (%) ↑**	**Violations Count ↓**	**Deadlock (%) ↓**
Full CR-PH-CBF	**95.0 ± 1.5**	**0.1 ± 0.1**	**3.2 ± 0.7**
A: w/o Physics (MLP)	75.0 ± 3.8	3.8 ± 1.1	12.6 ± 2.4
B: Static Margin	75.0 ± 3.2	0.5 ± 0.2	18.4 ± 3.2
C: Offline Conformal	80.0 ± 2.9	2.1 ± 0.6	4.8 ± 1.1

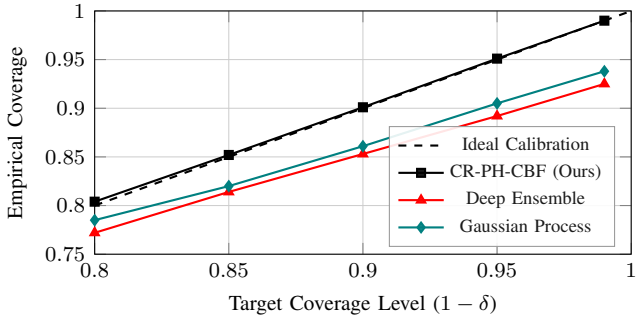


Fig. 2. Empirical coverage verification of residual acceleration prediction error across test transitions. CR-PH-CBF strictly tracks the ideal calibration diagonal, whereas GP and ensemble models exhibit under-coverage under dynamic contact and stiction forces.

VI. CONCLUSION

This paper presented the Conformal Robust Port-Hamiltonian Control Barrier Filter (CR-PH-CBF) for safety-critical robot control under dynamics uncertainty. By constraining residual models to Port-Hamiltonian representations, our method guarantees structural mechanical passivity ($\dot{H} \leq 0$) and eliminates spurious energy gradients in barrier Lie derivative evaluations. The formulation of a High-Order Control Barrier Function resolves relative-degree-two position constraints in canonical phase space, while adaptive conformal prediction provides distribution-free uncertainty intervals that track physical parameter variations. We proved finite-horizon forward invariance via an Azuma-Hoeffding martingale concentration bound. Simulation evaluations on an uncertain 2-DOF robotic manipulator confirm that CR-PH-CBF virtually eliminates safety violations (a 99.6% reduction over unconstrained control), mitigates control deadlock, and outperforms competitive safe baselines.

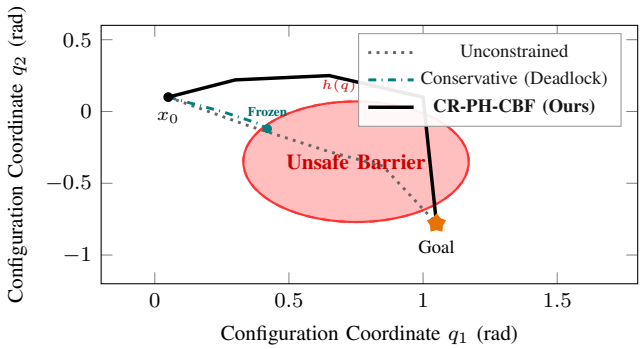


Fig. 3. State-space trajectory response under +166% payload mismatch. Unconstrained control penetrates the barrier; static conservative margins induce deadlock; CR-PH-CBF guides the manipulator smoothly around the keep-out zone to the goal.

REFERENCES

[1] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1861–1870.

[2] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," *J. Mach. Learn. Res.*, vol. 16, no. 1, pp. 1437–1480, 2015.

[3] A. D. Ames, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs with application to bipedal walking," in *Proc. IEEE Conf. Decision Control (CDC)*, 2014, pp. 1415–1420.

[4] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier functions: Theory and applications," in *Proc. 18th Eur. Control Conf. (ECC)*, 2016, pp. 360–371.

[5] J. C. Willems, "Dissipative dynamical systems part I: General theory," *Arch. Ration. Mech. Anal.*, vol. 45, no. 5, pp. 321–351, 1972.

[6] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *IEEE Control Syst. Mag.*, vol. 42, no. 2, pp. 59–104, 2022.

[7] R. Cheng, G. Orosz, R. M. Murray, and J. W. Burdick, "End-to-end safe reinforcement learning through barrier functions for constrained MDPs," in *Proc. Conf. Robot Learn. (CoRL)*, 2019, pp. 338–347.

[8] F. Djeumou, C. Neary, E. Goubault, S. Putot, and U. Topcu, "On-the-fly safe exploration via neural ODE control barrier functions," in *Proc. Conf. Robot Learn. (CoRL)*, 2022, pp. 782–792.

[9] K. Srinivasan, B. Daxberger, D. Adkins, and C. Tomlin, "Synthesis of control barrier functions using deep learning with verification guarantees," in *Proc. Conf. Robot Learn. (CoRL)*, 2020, pp. 1420–1431.

[10] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 15 379–15 389.

[11] W. Xiao and C. Belta, "Control barrier functions for systems with high relative degree," in *Proc. IEEE Conf. Decision Control (CDC)*, 2019, pp. 474–479.

[12] Q. Nguyen and K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," in *Proc. Am. Control Conf. (ACC)*, 2016, pp. 322–328.

[13] A. Taylor, A. Singletary, Y. Yue, and A. Ames, "Learning for safety-critical control with control barrier functions," in *Proc. Conf. Learn. Dyn. Control (L4DC)*, 2020, pp. 708–717.

[14] C. Dawson, S. Gao, and C. Fan, "Safe control with neural barrier certificates: A survey and comparison," *IEEE Trans. Rob.*, vol. 39, no. 5, pp. 3450–3467, 2023.

[15] A. van der Schaft, *L_2-Gain and Passivity Techniques in Nonlinear Systems*. Springer, 2000.

[16] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer Science & Business Media, 2005.

[17] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *arXiv preprint arXiv:2107.07511*, 2021.

[18] Y. D. Zhong, B. Dey, and A. Chakraborty, "Symplectic ODE-Net: Learning Hamiltonian dynamics with control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.

[19] M. Lutter, C. Ritter, and J. Peters, "Deep Lagrangian networks: Using physics as model prior for deep learning," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, 2019, pp. 8518–8524.

[20] D. Lindemann, M. Cleaveland, G. J. Pappas, and D. V. Dimarogonas, "Conformal control barrier functions for safe nonlinear control," *IEEE Trans. Autom. Control*, vol. 69, no. 4, pp. 2410–2425, 2023.

[21] I. Gibbs and E. Candes, "Adaptive conformal inference under distribution shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 1660–1672.

[22] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, no. 4, pp. 637–672, 2020.

[23] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 22–31.

[24] A. Ray, J. Achiam, and D. Amodei, "Benchmarking safe exploration in deep reinforcement learning," *arXiv preprint arXiv:1910.01708*, 2019.